Examiner only

8. *(a)* (i) Light of frequency less than $\frac{\phi}{h}$ cannot eject electrons from a surface of work function ϕ, even if the light intensity is increased. Explain this in terms of photons. [3]

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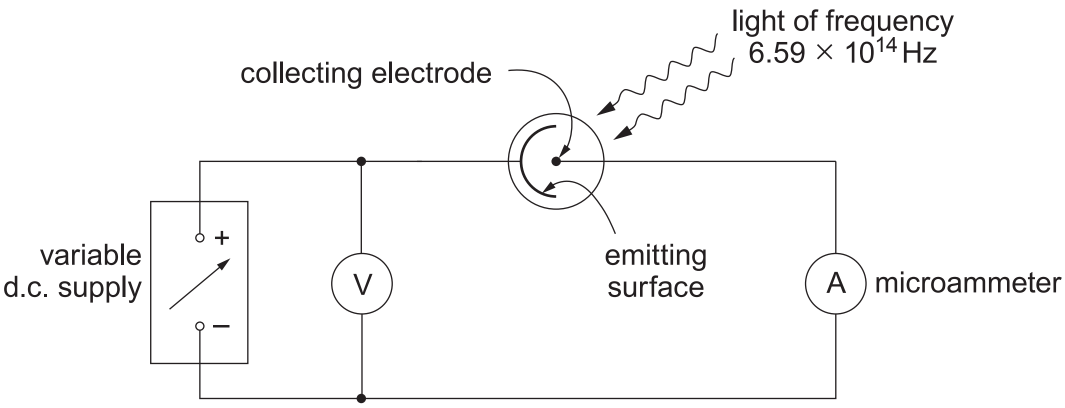
(ii) The emitting surface in a vacuum photocell is known to be made of one of the metals listed below (with their work functions).

Metal	caesium	potassium	barium	calcium	zinc
Work function / 10^{-19} J	3.12	3.68	4.03	4.59	5.81

The photocell is included in the circuit shown, and illuminated with light of frequency 6.59×10^{14} Hz.



With zero pd applied, the microammeter indicates a current. At some pd between 0 V and 0.35 V the microammeter reading drops to zero.

Examiner only

Determine which metal the emitting surface is made of, giving your reasoning clearly. [4]

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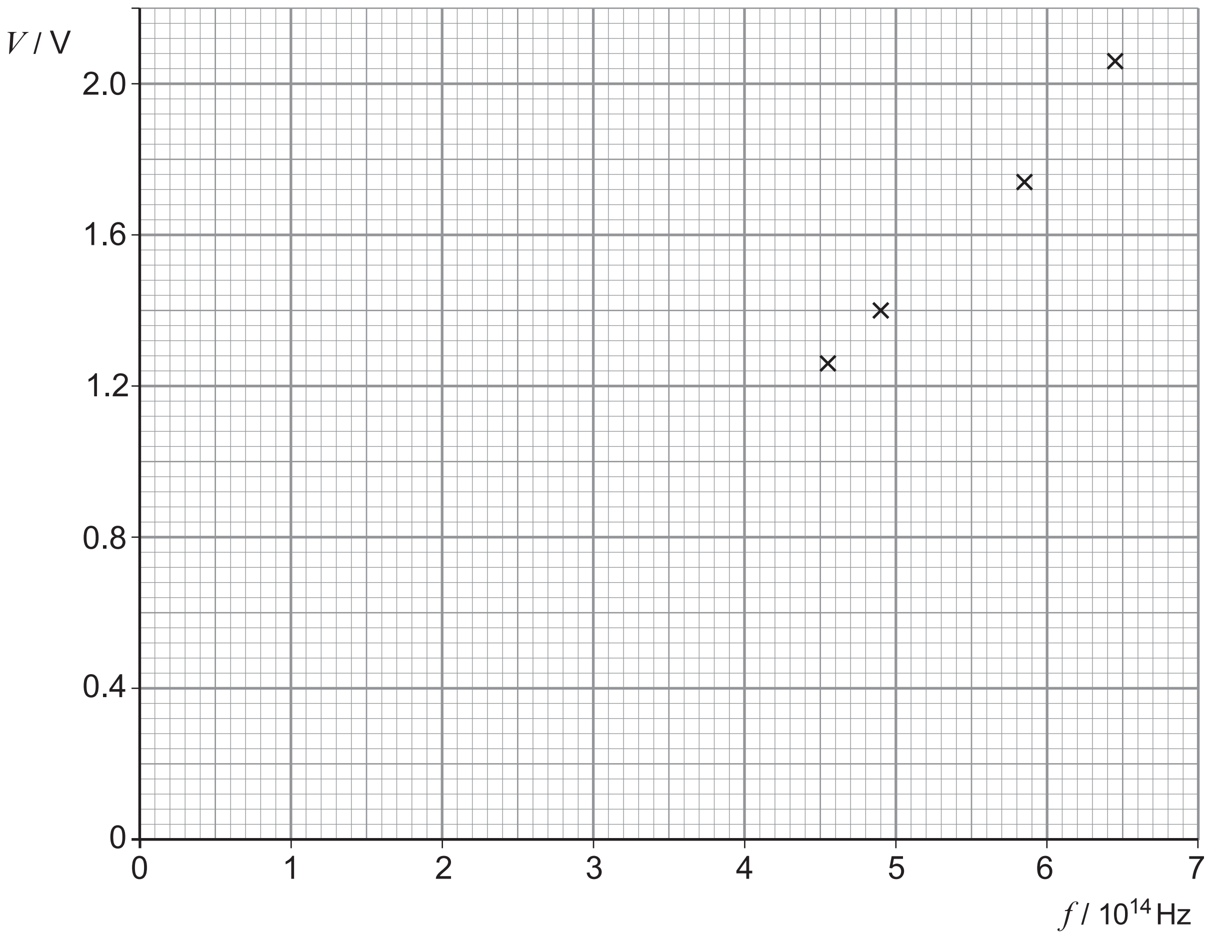
TURN OVER FOR THE REST OF THE QUESTION

Examiner only

(b) Rachel varies the pd across a light-emitting diode (LED) and notes the value, V, for which she can just see light from the LED. She also notes the frequency, f, of the light, as supplied by the LED's makers. She does the same for three other LEDs and plots V against f (below).



It has been suggested that V and f are related by the equation:

$$V = \frac{h}{e} f$$

(i) **On the graph** draw the line of best fit. [1]

(ii) Discuss the extent to which the graph supports an equation of this form. [2]

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Examiner only

(iii) Determine the gradient of the graph, and hence a value for h to an appropriate number of significant figures. Assume that the equation predicts $\frac{\Delta V}{\Delta f}$ correctly. [3] Show your working clearly.

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END OF PAPER